

Merge-Based Vocabulary Extension Stalls at the Pre-Tokenizer: Repairing Deployed LLMs for Nepali, and What It Costs

Anonymous ACL submission

Abstract

Llama-3, Qwen3 and many other open LLMs pre-tokenize text with a regular expression whose word class admits only Unicode letters. Devanagari vowel signs are combining marks, so this regex splits Nepali words at every vowel sign, and current open models that use it spend 2.6 to 5.8 times as many tokens on Nepali as on the same text in English. We ask what this means for adapting a model that is already deployed. Merge-based vocabulary extension cannot join the split pieces: for Llama-3 and Qwen3 no number of new merges brings the premium below about 2.2, and ten released extensions for six Indic and Southeast Asian scripts already end within 4% of this pre-token floor. We give a small edit to the deployed regex and prove that it changes no token id for text without combining marks or Devanagari characters. With 32,000 new merges it brings Nepali to 1.01 times English. We then ran a pre-registered comparison with continued pre-training of Llama-3.2-1B and Qwen3-0.6B on 1 GB of Nepali and 1 GB of English. Our main hypothesis failed: continued pretraining without new tokens gave the best held-out Nepali bits per byte, and the repaired model was worse by 0.066 (Llama) and 0.070 (Qwen). Against standard extension the repair was non-inferior for Llama (+0.006) but not shown to be for Qwen (+0.009). Scored after the separator used in training, a check we chose after seeing results, the Llama gap grows to +0.027. In exchange, the repaired model needs 2.2 times fewer decoding steps than standard extension for the same Nepali text. We release the tokenizers and code.

1 Introduction

On parallel FLORES-200 text, Qwen3’s tokenizer needs 4.42 times as many tokens for a Nepali sentence as for its English translation, and the c1100k tokenizer used by Phi-4, OLMo-2 and Granite-4.1 needs 4.76 times as many (§4). We call this ra-

tio the *premium*. Nepali speakers pay it on every request, in price, latency and usable context.

Much of the premium has a mechanical cause. GPT-2-style pre-tokenizers split text with a regular expression whose word class $\backslash p\{L\}$ matches one or more Unicode letters (OpenAI, 2019); we call it the *letters-only regex*. In Unicode, Devanagari vowel signs, the virama, the anusvara and the candrabindu are combining marks ($\backslash p\{M\}$), so the regex starts a new pre-token at each of them and byte-pair encoding (BPE) can never merge across that boundary. Regmi et al. (2026b) identified this cause and the resulting bound on token counts. They showed that adding marks to the word class cuts Nepali from 4.78 to 1.58 tokens per word, and that 63% of the 1,345 text-generation repositories they examined still use the letters-only class.

Regmi et al. studied tokenizers trained from scratch. Most practitioners instead adapt a model that already exists, usually by vocabulary extension: they add target-language tokens, initialise their embeddings and continue pretraining (Tejaswi et al., 2024; Yamaguchi et al., 2026; Purason et al., 2026; Smith et al., 2026). What is new in this paper is that deployed-model setting. We show that the bound becomes a floor that released merge-based extensions already hit; we give a repair of a deployed model’s regex that provably keeps English token ids; and we test it in controlled continued pretraining against the standard continued-BPE recipe of Purason et al. (2026), with the tokenizers released.

Findings. For Llama-3 and Qwen3 the *pre-token floor*, the lowest premium any merge-based extension can reach, is about 2.2 (§3). Ten released extensions for six scripts end within 4% of their floors, and our own extension under the letters-only regex stalls at it. Registering whole words as added tokens does get below the floor, but a fifth of those tokens then start in the mid-

dle of a word. Our repair adds marks to the word class with two string substitutions and, together with new merges, brings Nepali to parity with English while leaving every token id of mark-free, non-Devanagari text unchanged (§5). The model-quality result is less favourable. Under our pre-registered analysis, continued pretraining with no new tokens gave the best held-out Nepali bits per byte on both models, so our main hypothesis failed. Between the two extension recipes, the repair was non-inferior to standard extension for Llama and slightly worse for Qwen, and it was clearly worse for Llama when scored after the separator used in training (§7.2). It needs about half as many decoding steps as standard extension for the same Nepali text.

Contributions.

1. Evidence that the pre-token floor binds released merge-based extensions, and a test of the added-token route around it (§3); an audit of 24 distinct deployed tokenizers with per-tokenizer floors (§4).
2. An English-preserving repair of a deployed regex with a proof, a Devanagari-only variant with a stronger guarantee, and a practical pitfall in loading repaired tokenizers (§5).
3. A pre-registered comparison of R0 (no extension), R1 (standard extension) and R2 (repair + extension) in continued pretraining of two deployed models, reported against the registered hypotheses (§7.2).
4. The tokenizers, code and analysis plan.¹

2 Related work

Tokenization premiums. Tokenizers trained mostly on English encode other languages in more tokens, so those languages pay more for the same content (Rust et al., 2021; Petrov et al., 2023; Ahia et al., 2023; Lundin et al., 2026). For Nepali, Kadariya (2026) report 4.79 for cl100k, and Srivastava (2026) find that o200k cuts the mean Indic premium from 8.0 to 2.1 without attributing the change to the regex. We measure sentence-level token ratios on parallel FLORES-200 text (Goyal et al., 2022; NLLB Team et al., 2022), following Petrov et al. (2023). Fewer tokens do not guarantee a better model (Schmidt et al., 2024; Zouhar

et al., 2023), so we also test the repair with continued pretraining.

Pre-tokenization of combining marks. A 2024 tiktoken issue reported that GPT-2-style regexes split Indic words at vowel signs and proposed the class `[\p{L}\p{M}]` (ajaykg, 2024). ZeTT (Minixhofer et al., 2024) adjusted its regex to avoid over-segmenting Hindi and Tamil, Velayuthan and Sarveswaran (2025) showed that the GPT-2, GPT-4 and Llama-3 pre-tokenizers break Tamil, Sinhala and Hindi, and Rana et al. (2025) gained 38–40% in fertility by swapping the GPT-2 regex for Llama-4’s. Regmi et al. (2026b) identified the cause, stated the bound across 26 languages, and showed with 268M-parameter models trained from scratch that the wider class lowers Nepali bits per byte by 4.4%. Land (2026) reports that Qwen3.5 fixed the class. The edit itself is therefore known. What has not been studied is applying it to an already trained model while keeping its English tokenization.

Adapting the vocabulary of a pretrained model. Vocabulary extension adds tokens, initialises their embeddings from existing ones (Gee et al., 2022; Minixhofer et al., 2022; Dobler and de Melo, 2023; Liu et al., 2024) and continues pretraining (Tejaswi et al., 2024; Kim et al., 2024; Csaki et al., 2024; Nag et al., 2024); Duwal et al. (2024) continue pretraining Llama-3-8B on Nepali without new tokens, like our R0. Continued BPE appends new merges to a byte-level model’s merge list; Purason et al. (2026) introduce it and report tokenizer-only compression for many languages, Nepali included, under the unchanged regex. Yamaguchi et al. (2026) and Yamaguchi et al. (2025) extend Llama-3 and Qwen models for several abugidas, and Smith et al. (2026) expand LFM2. All of these keep the original pre-tokenizer. Tokenizer transfer (Minixhofer et al., 2024, 2025) and grapheme- or akshara-aware pre-tokenizers (Kapila and Bagavathy, 2026; Darshana, 2026) replace more of the tokenizer. Dagan et al. (2024) change the pre-tokenizer of Code Llama during fine-tuning and need more than 50B tokens to recover; their change alters English and code tokenization, which ours does not.

The closest prior system is TituLLM (Nahin et al., 2025), which extends Llama-3.2 for Bangla with a `\w`-based regex that keeps vowel signs attached. It does not identify the regex as a cause and has no regex-only or vocabulary-only arm,

¹Code, results, analysis plan and tokenizers: anonymised for review.

and its regex also changes digit and whitespace pre-tokens in English. We add a controlled attribution on deployed models and a repair that leaves every input without marks or Devanagari characters unchanged.

3 Pre-tokens bound merge-based extension

A byte-level BPE tokenizer (Sennrich et al., 2016; Radford et al., 2019) first splits a string s with a regex into pre-tokens $P(s)$ and then encodes each pre-token on its own by applying merges in rank order. Every pre-token yields at least one token, so $|T(s)| \geq |P(s)|$; Regmi et al. (2026b) call this the pre-tokenization ceiling. Appending merges, whatever they are and however many, changes neither the normalizer nor the regex, so the bound holds for every tokenizer extended this way.

We turn this into a bound on the premium. For a tokenizer T with pre-tokenizer P , the *pre-token floor* is $|P(\text{ne})|/|T(\text{en})|$, the number of Nepali pre-tokens divided by the number of English tokens, summed over the 1,012 FLORES-200 devtest sentences. A merge-based extension that keeps P and leaves English unchanged cannot bring the premium below this value, though it may stop above it. Our standard extension R1 leaves English unchanged: every new merge produces a token that contains a complete Devanagari code point, and since UTF-8 is self-synchronising, no new merge can fire on text without one (the same argument appears in the proof of Proposition 1, Appendix C).

For today’s regexes the floor is far from slack. In Llama-3 and Qwen the word branch is $[\text{^\wedge r \n \p{L} \p{N}}]? \text{ \p{L}+}$, a run of letters optionally preceded by one other character. A Devanagari vowel sign ends the current run and becomes the leading character of the next one, so *nepālko sarkār* (“government of Nepal”) becomes six pre-tokens, four of which start with a vowel sign that belongs to the previous consonant (Figure 4, Appendix J). The floor is 2.19 for Llama-3 and 2.17 for Qwen3. That is above the premium of o200k (1.61), and 2.7 to 2.8 times the floor under the repaired regex of §5 (0.79 and 0.80). Llama-3’s premium is already 2.58, so standard extension can lower it by at most 15%.

Released extensions end at the floor. Figure 1 measures released merge-based extensions of letters-only models; Appendix I lists every number and how we found them. Ten extensions by

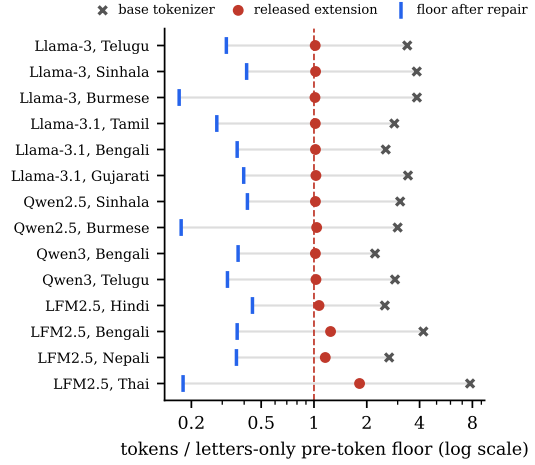


Figure 1: Released merge-based extensions of letters-only models on FLORES-200 devtest of each target language, relative to the letters-only pre-token floor. The ten extensions by Yamaguchi et al. (top) move from the base tokenizer (x) to within 4% of the floor (●). LFM2.5 (bottom) stops further above it for Bengali and Thai. The blue tick is the floor under the repaired regex. Full numbers, including the Amharic control, are in Table 6.

Yamaguchi et al. of Llama-3, Llama-3.1, Qwen2.5 and Qwen3 (Yamaguchi et al., 2026, 2025) cover six scripts written with combining marks. All ten end within 4% of their floor, having closed at least 98% of the distance from the base tokenizer.² The 61k-token in-place expansion of LFM2 (Smith et al., 2026) ends further from its floor (1.07 times it for Hindi, 1.16 for Nepali, 1.24 for Bengali and 1.82 for Thai, which is written without spaces), but its floors are just as high. Under the repaired regex the same text has 2.2 to 5.9 times fewer pre-tokens, so each of these extensions stopped well short of what new vocabulary could achieve. Yamaguchi et al.’s Amharic extension, for a script without combining marks, stays at 1.57 times its floor, and the repair does not change that floor.

Added tokens bypass the floor. Strings registered as Hugging Face *added tokens* are matched in the raw text before the regex runs, so the bound does not apply to them. SinLlama (Aravinda et al., 2025) extends Llama-3 for Sinhala this way and goes below its letters-only floor. To test this route fairly, we registered as added tokens the 31,753 whole Devanagari strings among

²These extensions re-order the base merges and change English token counts slightly (+6.5% for the Llama-3 Telugu release), so for them we compare tokens with pre-tokens of the target text directly.

the 32,000 learned for R2 (§5), on the unchanged regex. Nepali then costs 1.025 times English on Llama-3 and 1.009 on Qwen3, close to R2’s 1.014, and no FLORES-200 sentence without a Devanagari character changes its token ids. The floor therefore binds merge-based extension only. The price is the segmentation. Added tokens match the longest string anywhere, ignoring word boundaries, and 21% of the matches start inside a word and 28% end inside one. Allowing only whole-word matches (`single_word=True`) removes those splits and also the gain: the premium returns to 2.49 on Llama-3 and 4.22 on Qwen3 (Appendix J). We did not train models with added tokens.

4 Audit of deployed tokenizers

We measured 33 released tokenizers on FLORES-200 devtest. Nine produce token ids identical to another tokenizer’s on every sentence of five languages we checked (for example, Phi-4, OLMo-2 and Granite-4.1 match c1100k, and Qwen3 matches Qwen2.5), which leaves 24 distinct tokenizers (Figure 2, Appendix F). We classify each by its own pre-tokenizer configuration, since models in one family do not always share a regex. Four SentencePiece and WordPiece tokenizers (XLM-R, mT5, NLLB-200, IndicBERTv2) do not round-trip their input, so their low premiums are not comparable; we mark them and leave them out.

Eight distinct tokenizers (13 released models, including GLM-5.3 and LFM2) use the letters-only class, and none of them can be extended below 2.15. Leaving aside GPT-2 (7.55), which we include only for reference, their premiums range from 2.58 (Llama-3) to 5.77 (LFM2). The seven lossless tokenizers with a mark-aware regex range from 1.12 (Arkios; Regmi et al., 2026a) to 2.97 (DeepSeek-V3).

We split each premium into three factors (Appendix G). The base term, 0.77 to 0.80 for every tokenizer, is what the premium would be if every Nepali pre-token under the repaired regex were a single token. The regex factor, 2.77 to 2.79 for the Llama-3, Qwen and c1100k pattern, counts the extra pre-tokens the tokenizer’s own regex creates. The vocabulary factor, tokens per pre-token, is the only one that merge-based extension can reduce, and it cannot go below 1. For Llama-3 it is already 1.18, so most of the remaining premium sits in the regex factor. The factors are an accounting iden-

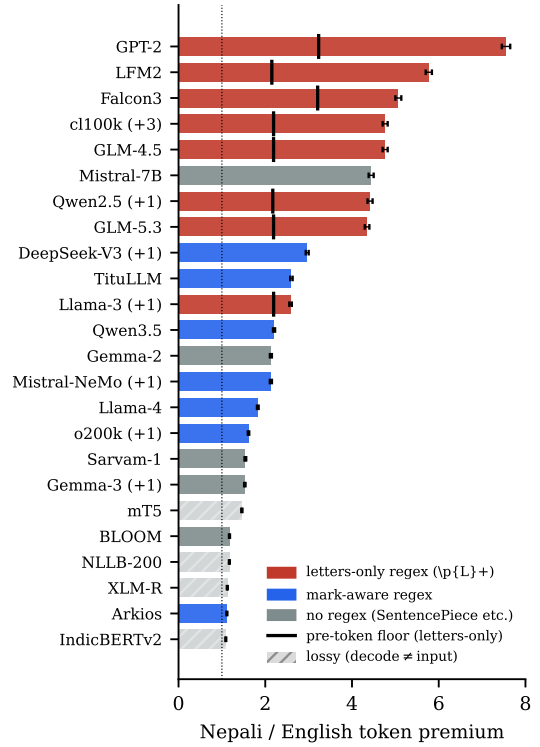


Figure 2: Nepali/English premium of 24 distinct tokenizers (33 released; “+n” counts aliases) on FLORES-200 devtest, with 95% paired-bootstrap intervals. Black ticks mark the pre-token floor of each letters-only tokenizer: no merge-based extension can move its bar to the left of the tick.

tity. The regex factor measures room for improvement, and repairing the regex alone does not lower the premium (§7.1). Among mark-aware tokenizers the vocabulary factor still ranges from 1.43 (Arkios) to 3.79 (DeepSeek-V3), so a good regex also needs Nepali merges. Within families, the Nepali premium halves from Qwen3 to Qwen3.5 (4.42 to 2.20), whose word branch is close to our repaired class (Land, 2026), and falls from 2.58 to 1.83 from Llama-3 to Llama-4, which adopts o200k’s pattern. Both vocabularies also grew, so these changes are consistent with our account without isolating the regex.

A well-formed Devanagari syllable never starts with a combining mark, yet 55% of Llama-3’s Nepali tokens do. Even o200k’s mark-aware regex leaves 26%, because BPE also splits inside words.

5 Repairing a deployed tokenizer

The repair has three parts. We edit the regex, learn new merges on top of the existing ones, and initialise the new embeddings so that the model behaves exactly as before on old tokens.

Regex edit. We add $\backslash p\{M\}$ to the word class and exclude it from the optional leading character and from the punctuation class. A mark can then only be matched as part of a word, so it stays with the letters around it. For the Llama-3 and Qwen pattern this takes two string substitutions that edit three character classes (Appendix B); digits, contractions, whitespace and punctuation are handled as before. We call the result the *repaired regex*. Qwen3.5 ships almost the same word branch (Land, 2026).

Continued BPE. Following Purason et al. (2026), we pre-tokenize Nepali text with the arm’s regex, segment every pre-token with the original merges, as the deployed model sees it, and run BPE over these segmentations with each base token as an indivisible symbol. The K new merges are appended after the existing ones, so the original merges act first. We discard any merge whose output contains no code point in U+0900–U+097F, together with merges that depend on it. R1 and R2 run this procedure on the same text with the same K and differ only in the regex.

Proposition 1 (English invariance). *Let s be a normalised string that contains no combining mark ($\backslash p\{M\}$) and no code point in U+0900–U+097F. Then the repaired tokenizer R2 and the original tokenizer produce identical token ids for s .*

The proof is in Appendix C. The proposition covers ASCII English and code and every script whose normalised text has no combining marks. It does not cover other scripts that use marks: the general repair changes the pre-tokens of 61 of the 204 FLORES-200 languages, including Arabic, Bengali and Thai, whose tokenization the base model has seen only in its original form.

A Devanagari-only repair. Admitting only the 34 combining marks of the Devanagari block, instead of all of $\backslash p\{M\}$, gives a stronger guarantee by the same argument: token ids are unchanged for every input without a code point in U+0900–U+097F. On FLORES-200 this variant changes pre-tokens only in the ten Devanagari languages, and on FLORES Nepali its R2 token ids are identical to the general repair’s. On a 275 MB sample of FineWeb-2 Nepali, 24 of 49,462 lines are pre-tokenized differently, all because of marks from other blocks, such as emoji variation selectors and Kannada vowel signs (Appendix B). We recommend this variant for deployment. All experiments

in this paper, and the released tokenizers, use the general repair.

Embedding initialisation. Each new token’s input and output rows are set to the mean of the rows of its constituent base tokens (Gee et al., 2022), obtained by applying the original merges to its byte-level string. Original rows are untouched, so on any input whose ids are all original the resized model gives the same logits for every original token as the base model; on an English probe the largest difference is below 10^{-4} . The new tokens do add terms to the softmax denominator, and continued pretraining is needed to train them.

Loading pitfall. The transformers class Qwen2Tokenizer rebuilds the pre-tokenizer from a hard-coded pattern when it loads, which silently undoes a repaired regex stored in tokenizer.json. Our released tokenizers therefore set tokenizer_class to PreTrainedTokenizerFast, and we check after loading that the premium matches the paper.

Arms. R0 (no extension) keeps the original tokenizer. R1 (standard extension) appends K merges learned under the original regex, as in Purason et al. (2026) and Yamaguchi et al. (2026). R2 (repair + extension) switches to the repaired regex and appends K merges learned under it.

6 Experimental setup

We registered the retrofit study in the project repository before training any model. A first amendment, committed before any continued-pretraining result, recorded our budget cuts and fixed the primary analysis. A second amendment, written after the first Llama results, added the warm-up runs described below. Appendix A gives all four commits and every deviation; details of training and evaluation are in Appendix D.

Models and data. We use Llama-3.2-1B (base) and Qwen3-0.6B-Base, whose tokenizers ship the letters-only regex and which tie input and output embeddings. The Qwen3 tokenizer is identical across model sizes; we built it on Qwen3-1.7B-Base. Nepali text is FineWeb-2 np_i_Deva and English is FineWeb-Edu. We normalise to NFC, deduplicate, and remove every document that shares a whitespace 10-gram with FLORES-200 dev or devtest. A 30 MB held-out slice of each language is never trained on. R1 and R2 learn

merges on the same 1 GB of Nepali, with the registered $K=32,000$.

Continued pretraining. Each arm is trained for one epoch on the same documents, 1 GB of Nepali and 1 GB of English interleaved in one fixed order, with identical optimisation: a fixed, untuned peak learning rate of 10^{-4} , cosine decay and one run per arm. Because R2 encodes the same bytes in fewer tokens, it takes fewer steps. R0 and R1 also save a checkpoint at the step where R2 ends, before their learning rate has decayed. We call these the *main runs*. The *warm-up runs* (Llama only) repeat them after first training only the embedding matrix on 10% of the stream at a learning rate of 10^{-3} .

Evaluation. All metrics are normalised by UTF-8 bytes. The primary metric is bits per byte (BPB) on held-out native Nepali, *byte-matched*: each document is split at whitespace into pieces of about 2,000 bytes, and every arm scores the same pieces. Each piece is scored after one context token. The *registered prefix* is the beginning-of-sequence (BOS) token for Llama and the end-of-text token for Qwen, which has no BOS. During training, however, documents were separated only by the end-of-text token, so the Llama models never saw BOS. We therefore also score Llama after that token, the *training separator*, as a diagnostic chosen after seeing results (§7.2). Secondary metrics (held-out English BPB, FLORES BPB, Belebele, 5-shot translation) are in Appendix K.

Analysis. The registered main hypothesis was that R2 is non-inferior to both R1 and R0 on held-out Nepali BPB, with a margin of 0.01, while using fewer training tokens. The margin is 1.3% of the base Llama model’s Nepali BPB and 1.2% of base Qwen’s. We call R2 better if the 95% interval of (R2 – other) lies entirely below 0 and non-inferior if it lies entirely below +0.01. The first amendment called every other case “no detectable difference”; the second split off “worse” for an interval entirely above 0, and we use that label because the first would misdescribe such an interval. The registered test was a paired bootstrap over held-out pieces. Pieces of one document are correlated, so we report a bootstrap over documents instead ($B=10,000$). Document ids were not stored with the pieces; we reconstruct them by taking a piece shorter than 1,900 bytes to end a document, which gives 484 Nepali documents from 2,259 pieces. On Nepali this widens the in-

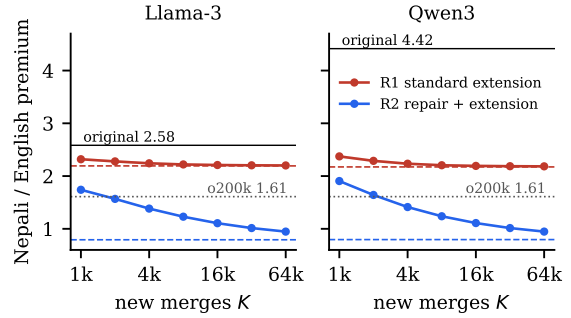


Figure 3: Nepali/English premium on FLORES-200 devtest after adding K merges by continued BPE, under the letters-only regex (R1) and the repaired regex (R2). Dashed lines: pre-token floors under each model’s own pattern (2.19 and 2.17 letters-only; 0.79 and 0.80 repaired, for Llama-3 and Qwen3). Shading: 95% paired-bootstrap intervals.

tervals by a factor of 1.3 to 2.3 and changes no verdict. The same test on English pieces is the registered English control. All intervals describe the sampling of evaluation text. With one run per arm, they say nothing about run-to-run variance.

7 Results

7.1 Compression

Figure 3 sweeps the number of new merges K . Under the letters-only regex (R1), extension saturates almost at once. For Llama-3 the first 1,000 merges take the premium from 2.58 to 2.32, and the next 63,000 only to 2.20, within 0.01 of the floor. For Qwen3, R1 reaches 2.19 by $K=16k$ and stops at 2.18, against a floor of 2.17. It stops early because the letters-only regex leaves few distinct Nepali pre-tokens to merge (Appendix J). Under the repaired regex (R2), additional merges keep helping. At $K=32k$ Nepali reaches parity with English (1.01) on both models, and at 64k it is cheaper than English (0.95), below o200k. At $K=32k$, R2 needs 46% of R1’s tokens for FLORES Nepali. All 28 extended tokenizers decode losslessly and reproduce the original token ids on all 1,012 English devtest sentences, as §3 and Proposition 1 predict.

Both parts of the repair are needed. Repairing the regex without new merges changes almost nothing (2.61 for Llama-3, 4.42 for Qwen3), because the old merges still cut Nepali words into the old pieces. The repair also mends syllables. The share of Nepali tokens that start with a combining mark rises from 55% (Llama-3) and 32% (Qwen3)

	Llama-3.2-1B			Qwen3-0.6B
	main runs		warm-up runs	main runs
	registered prefix	training separator	training separator	(both identical)
<i>Held-out Nepali BPB</i> (lower is better)				
Base model, no training	0.766	n/a	n/a	0.852
R0	0.474	0.471	0.471	0.501
R1	0.534	0.483	0.480	0.561
R2	0.539	0.510	0.501	0.570
<i>Paired difference in held-out Nepali BPB</i> (positive: first arm worse), document-clustered 95% interval				
R1 – R0	+0.060 [+0.058, +0.061]	+0.012 [+0.011, +0.013]	+0.010 [+0.009, +0.011]	+0.060 [+0.058, +0.062]
R2 – R0	+0.066 [+0.064, +0.068]	+0.039 [+0.038, +0.041]	+0.031 [+0.029, +0.032]	+0.070 [+0.067, +0.072]
R2 – R1	+0.006 [+0.004, +0.007]	+0.027 [+0.026, +0.028]	+0.021 [+0.020, +0.022]	+0.009 [+0.008, +0.010]
Training tokens (M), R0/R1/R2		417 / 386 / 286		574 / 392 / 292
Decoding steps, relative to R2		2.55 / 2.17 / 1		4.35 / 2.16 / 1

Table 1: Continued pretraining on the same 1 GB of Nepali and 1 GB of English in every arm (one run per arm). BPB: bits per byte on held-out native Nepali, scored in pieces of about 2,000 bytes. Registered prefix: the beginning-of-sequence token, as pre-registered. Training separator: the end-of-text token that preceded every document during training (a post hoc diagnostic). Qwen has no beginning-of-sequence token, so both prefixes are the end-of-text token. Intervals resample 484 documents (2,259 pieces; a piece shorter than 1,900 bytes is taken to end a document) and reflect only the sampling of evaluation text. Warm-up runs under the registered prefix and all secondary metrics are in Table 8. Decoding steps: tokens needed for FLORES-200 Nepali devtest, relative to R2.

to 65% and 64% under R1, since new merges can only build on mark-initial fragments, and falls to 13% and 11% under R2.

We repeated the sweep for seven other languages written with combining marks and two controls (Table 5, Appendix H), learning merges on 25 to 361 MB of FineWeb-2 text per language. At $K=32k$, R2 brings all seven to between 0.88 and 1.24 times English. R1 ends within 5% of its floor in six of them. Thai is the exception: it is written without spaces, so its letters-only pre-tokens are long and leave R1 room. For Amharic and Korean, whose scripts have no combining marks, R1 and R2 are identical at every K , as the mechanism predicts.

7.2 Continued pretraining

The registered outcome. Table 1 gives the results. The main hypothesis fails against R0. Continued pretraining without new tokens gave the lowest held-out Nepali BPB on both models, and R2 was worse than R0 by 0.066 on Llama and 0.070 on Qwen, far beyond the 0.01 margin. Under this prefix standard extension also lost to R0, by 0.060 on both models. The new embeddings start as averages of existing ones, and 1 GB of Nepali is not enough for them to catch up with tokens the base model already knows. Against

standard extension, the registered comparison gives $R2 - R1 = +0.0058$ [$+0.0044, +0.0073$] for Llama, which is non-inferior, and $+0.0094$ [$+0.0084, +0.0104$] for Qwen. The Qwen interval excludes zero and its upper end crosses the margin, so non-inferiority is not shown and R2 is worse. That verdict turns on 0.0004 BPB, a difference a second training run could plausibly move; we did not measure run-to-run variance.

The training separator. For Llama, scoring after the training separator instead of BOS lowers BPB by 0.003 for R0, 0.051 for R1 and 0.029 for R2, so the choice of first token changes the comparison. After the separator, R2 is worse than R1 by $+0.027$ [$+0.026, +0.028$]. We consider the separator scores closer to how the models were trained, but we chose them after seeing results, and BOS-prefixed inference is the default in Hugging Face and in chat templates, so the registered numbers are what a deployment would see. We have not established why the shift depends on the arm. With tied embeddings, the BOS row is updated as an output class even though it never appears as input, and in the warm-up runs its norm grows from 0.863 to 0.896 (Appendix K); this may contribute. Training with BOS at every document start would remove the problem.

What the repair buys. R2 trains on 26% fewer tokens than R1 for the same bytes, but this does not make it cheaper to reach a given quality. At R2’s final step count, the R0 checkpoint, whose learning rate had not yet decayed, is already better than the final R2 model by 0.060 [0.056, 0.063] on Llama and 0.053 [0.049, 0.057] on Qwen. The R1 checkpoint at that step is better by 0.006 [0.003, 0.008] on Llama and level on Qwen (scored on the first 1 MB of held-out text, registered prefix). Measured training time fell by only 3% (Llama) and 11% (Qwen) relative to R1, in runs that shared GPUs with evaluation jobs. The clear gain is at inference: for the same FLORES Nepali text, R2 needs 2.17 (Llama) and 2.16 (Qwen) times fewer decoding steps than R1, which has the same vocabulary size, and 2.55 and 4.35 times fewer than R0.

Warm-up runs. Training only the embedding matrix first leaves R0 where it was (0.471 after the separator) and improves R2 from 0.510 to 0.501. R2’s deficit relative to R0 shrinks from 0.039 to 0.031, as the second amendment predicted, and its gap to R1 from 0.027 to 0.021 [0.020, 0.022]. These are differences between single runs, so we treat them as suggestive. The amendment stated its prediction for the registered prefix, where the warm-up makes Llama scores unusable: R0 moves from 0.474 to 0.779.

English. The registered English control holds. R2 – R0 on held-out English is +0.0003 [–0.0002, +0.0008] for Llama (non-inferior) and –0.0028 [–0.0032, –0.0024] for Qwen (better), and no two arms of a model differ by more than 0.004. This is expected, since the tokenizer change does not touch English and every arm sees the same English data. Under the registered prefix all three Llama arms are slightly worse on English than the base model (0.799 to 0.803 against 0.788).

8 Discussion

Recommendations. Before extending the vocabulary of a letters-only model (Table 3) for a language with combining marks, count the target text’s pre-tokens under the model’s regex and compare them with its tokens; our released code includes a one-command check. If the two are close, as for Llama-3 on Nepali, merge-based extension has little room. At our budget, plain continued

pretraining gave the best Nepali quality, so extension of either kind needs a reason beyond quality. If inference cost is that reason, the repair roughly halves the decoding steps of standard extension at a cost in BPB of 0.006 to 0.027 on Llama and 0.009 on Qwen. We would use the Devanagari-only variant, which leaves every other script untouched, and load the tokenizer with a class that does not rebuild its pre-tokenizer. Continued-pretraining recipes should keep the BOS token at document starts, or evaluate after the separator the model was trained with; otherwise the choice of first token can change comparisons between models.

What the repair does not fix. A wider word class removes the splits that the regex forces, but BPE can still split inside a syllable: 26% of o200k’s Nepali tokens and 11 to 13% of R2’s start with a combining mark. An akshara-aware rule (Kapila and Bagavathy, 2026; Darshana, 2026) restricted to the Devanagari block would keep English token ids by the argument of Proposition 1; testing one in a deployed model is future work. Added tokens are a second route past the floor that needs no regex change, but at the cost of the segmentation (§3), and we have not measured their effect on model quality.

Limitations

- **The main hypothesis failed.** At our budget, neither extension recipe beat continued pretraining without new tokens on Nepali BPB, and the repair was not shown non-inferior to standard extension for Qwen. Our evidence for the repair comes from compression and decoding steps only.
- **Scale.** We adapt 1B- and 0.6B-parameter models with 1 GB of text per language and one epoch. The registered 3 GB runs were dropped for budget reasons, so we cannot say whether the gap between arms closes with more data; Dagan et al. (2024) needed more than 50B tokens to recover from a pre-tokenizer change.
- **One run per arm.** Every interval reflects only the sampling of evaluation text. We did not measure run-to-run variance, which is probably larger. Differences of a few thousandths, including the Qwen non-inferiority verdict and the warm-up effect, could change

with another seed.

- **Approximate clustering.** Document boundaries in the evaluation set are reconstructed from piece lengths, so the clustered intervals are approximate.
- **Untuned optimisation.** One fixed learning rate (10^{-4}) is used for every arm; the registered learning-rate selection was not run. R2 takes fewer, larger steps than R1 and relies more on new embedding rows, so this recipe may handicap it. Mean initialisation may also suit R2’s long new tokens less well than R1’s short ones.
- **No BOS in training.** Our continued-pretraining data never placed Llama’s BOS token at document starts, yet the registered evaluation puts it before every input. This makes the Llama comparison depend on a scoring choice, and our separator diagnostic was chosen after seeing results. All Llama secondary metrics also use BOS, so the warm-up runs’ translation scores are unusable.
- **Weak secondary metrics.** Belebele accuracy is near chance for every model, translation scores come from a single greedy pass without intervals, and measured generation speed was taken on shared GPUs (Appendix K).
- **Equal compute.** The equal-compute checkpoints of R0 and R1 are taken before their learning rate has decayed, and are scored on 1 MB only.
- **Languages.** We train models for Nepali only. For other languages we report compression, and the general repair changes the tokenization of 61 FLORES-200 languages whose quality we did not measure. The cross-script tokenizers use a relaxed merge filter not covered by Proposition 1 (Appendix H).
- **Added tokens.** We measured the added-token route around the floor only at the tokenizer level and trained no model with it.
- **What would settle it.** Retraining with BOS at document starts, three seeds per arm, a learning-rate sweep, annealed equal-compute runs and a curve over data budgets would show whether the gap between R2 and R1 comes from the tokenizer or from this recipe.

Ethics and use of AI assistance

All text and evaluation data are public (FineWeb-2, FineWeb-Edu, FLORES-200, Belebele) and used under their licences; we collected no data from people. Released Llama-derived tokenizers follow the Llama 3.2 Community License (Meta Platforms, Inc., 2024); Qwen3-derived tokenizers are Apache-2.0.

We used Claude (Anthropic), an AI system, extensively. Under our direction it wrote and ran the experiment code, searched the literature and drafted the text. The authors reviewed and approved the study design, the paper and all citations, and take responsibility for the content. The numbers in the paper can be reproduced from public tokenizers and data with our released code.

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	A Pre-registration history	975
	We committed the analysis plan to the project repository in four versions, all on 2026-10-04, and report all four. Times are commit times.	976
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1. **Original registration** (bf5ebe4, 17:59) registered hypotheses about the mechanism, a regex-only tokenizer ablation, a cross-script dose-response and training from scratch. Our literature check then found that [Regmi et al. \(2026b\)](#) had published the mechanism, the ablation, the cross-script analysis and a from-scratch comparison five weeks earlier. We withdrew those hypotheses as contributions.
2. **Retrofit registration** (3c03367, 18:16), before any model was trained or evaluated, registered the study reported here: arms R0, R1, R2 and the untrained base model; $K=32,000$ unless R2 saturated earlier on FLORES dev; equal-bytes continued pretraining of Llama-3.2-1B and Qwen3-1.7B-Base on 3 GB of each language; a learning rate selected once per model on R0; two seeds. Its hypotheses were P1 (compression ordering, with R1 above the letters-only floor), P2, the main hypothesis (held-out Nepali BPB of R2 at most 0.01 above both R1 and R0, while using fewer training tokens), P3 (English BPB of R2 and R0 within 0.01) and P4 (R2 decodes Nepali faster at equal bytes). P1 had already been measured when it was written, so we report it as a measurement.
3. **First amendment** (5ff74e8, 19:50), written after an internal review and before any continued-pretraining run had produced an evaluation, recorded the deviations below, fixed four pipeline bugs, and fixed the primary analysis: byte-matched BPB, a paired bootstrap over pieces and the claim rules of §6. Earlier launches had been stopped by us because of the bugs, and none produced an evaluated checkpoint. The only earlier evaluation, of the untouched base Llama-3.2-1B with a since-fixed prefix bug, is discarded.
4. **Second amendment** (bfab80d, 21:46) was written *after* observing the Llama R0 and R2 main-run results and before any R1 or Qwen result. It registered the warm-up runs, with the prediction that they shrink the R2 – R0 gap, and 3 GB runs that we did not run, and it added the “worse” category to the claim rules.

In the repository these are PREREG.md, PREREG_RETROFIT.md, PREREG_AMENDMENT.md and PREREG_AMENDMENT3.md. All four commits

lie within four hours in a repository we control, so they document our plan without an external timestamp.

Outcome of each registered hypothesis. P1 holds as a measurement (§7.1). P2 fails against R0 on both models; against R1 it holds for Llama and is not shown for Qwen (§7.2). P3 holds. P4 holds for the deterministic number of decoding steps; our measured throughput is too noisy to test it (Appendix K). The warm-up prediction holds after the training separator; under the registered prefix the warm-up scores are dominated by the BOS artifact.

Deviations from the retrofit registration. The first amendment recorded these before any result:

- Qwen3-0.6B-Base replaces Qwen3-1.7B-Base, for budget and time. Their tokenizers are identical.
- 1 GB of Nepali and 1 GB of English per arm replace 3 GB each, with identical documents in every arm.
- A fixed learning rate of 10^{-4} replaces selection on R0 from $\{3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}\}$ by a 200-step pilot, which was never completed.
- One seed per arm replaces two.
- Added: R0 and R1 checkpoints at R2’s final step (equal-compute view); byte-matched BPB as the primary metric, with the registered sliding-window BPB as secondary; the paired bootstrap with claim rules as the primary test; a translation cap of 512 instead of 256 new tokens.

After seeing results we made four further changes, all reported as such: intervals from a bootstrap over reconstructed documents instead of pieces (§6); the training-separator diagnostic; decoding steps instead of measured bytes per second for P4, whose implementation had fixed the number of generated tokens rather than bytes; and the “worse” label from the second amendment for the main runs.

Pipeline bugs fixed before any reported run.

(1) Documents had been stored one per line with their internal newlines, so “documents” were paragraphs; they are now stored as one JSON-encoded document per line. (2) New-token embeddings had been initialised by decoding tokens to text,

which turns partial UTF-8 tokens into U+FFFD; constituents now come from applying the original merges to the byte-level string. (3) The initialisation check had covered Qwen’s padding rows; it now covers only the original vocabulary. (4) Evaluation had prefixed documents with the end-of-text token, while base Llama expects BOS; the registered evaluation uses BOS where it exists.

B Repair details

Regex edit. For the Llama-3 and Qwen pattern we make two string substitutions. The first replaces $\backslash p\{L\}^+$ by $[\backslash p\{L\}\backslash p\{M\}]^+$ in the word branch. The second inserts $\backslash p\{M\}$ after $\backslash p\{L\}$ in the two negated classes, the optional leading character of the word branch and the punctuation branch. Three character classes change:

$$\begin{aligned} & [\backslash r\backslash n\backslash p\{L\}\backslash p\{N\}]^?\backslash p\{L\}^+ \\ & \quad \downarrow \\ & [\backslash r\backslash n\backslash p\{L\}\backslash p\{M\}\backslash p\{N\}]^?[\backslash p\{L\}\backslash p\{M\}]^+ \\ & _?[\backslash s\backslash p\{L\}\backslash p\{N\}]^+ \rightarrow _?[\backslash s\backslash p\{L\}\backslash p\{M\}\backslash p\{N\}]^+ \end{aligned}$$

Nothing else in the pattern changes.

Devanagari-only variant. The variant replaces $\backslash p\{M\}$ in the three classes by an explicit class of the 34 combining marks of the Devanagari block (U+0900–0903, U+093A–093C, U+093E–094F, U+0951–0957, U+0962–0963). We checked that this class equals the set of $\backslash p\{M\}$ code points in the block under Python’s unicodedata (Unicode 15.0), the regex module and the Hugging Face tokenizers engine. Every other character has the same class membership as in the original regex, so any text without a Devanagari mark is pre-tokenized exactly as before. Across the 204 FLORES-200 devtest files, the general repair changes pre-tokens in 61 languages for both models and the Devanagari-only variant in exactly the ten Devanagari languages. On FLORES Nepali the two variants give identical pre-tokens and identical R2 token ids. On a 275 MB sample of FineWeb-2 Nepali (49,462 lines), 24 lines differ, in 47 segments; the marks involved are emoji variation selectors, Kannada, Bengali, Thai and Hebrew signs and one combining candrabindu in romanised text, and the R2 token count of those lines is the same under both.

Continued BPE. At inference BPE applies the lowest-ranked applicable merge first, so the original merges act before the new ones. The extended tokenizer reproduces the training procedure except in two cases: a new merge whose output string is

already in the original vocabulary can let an original merge fire again afterwards, and under Hugging Face’s `ignore_merges` option (set for Llama-3) a pre-token that is itself a vocabulary entry is emitted directly. Neither case affects Proposition 1. We run merge learning inside the Hugging Face Rust trainer by mapping each base token to a private-use code point; learning $K=32,000$ merges on 1 GB of Nepali takes 1.7 to 3.4 minutes per tokenizer.

Scope of Proposition 1. The proposition covers ASCII English and code and every script whose normalised text has no combining marks, including most Cyrillic, Han, Hangul and Ethiopic text. It does not cover Devanagari, which changes by design, or text with marks, such as Hindi, Arabic with vowel points, Thai, or Latin written in decomposed form (NFD). The Llama-3 tokenizer has no normalizer, so deployments should apply NFC first; Qwen3’s tokenizer applies NFC itself. It also assumes that the regex engine’s $\backslash p\{M\}$ table matches the one used to define s ; for the Devanagari-only variant we checked this directly. Empirically, R1 and R2 at $K=32,000$ give token ids identical to the original tokenizer on all 1,012 FLORES devtest sentences in English, Russian and Amharic, for both models, including one Russian sentence with a combining acute that lies outside the proposition.

C Proof of Proposition 1

Proof. Pre-tokens. The repaired regex differs from the original only in the three classes of Appendix B, and each differs from its original only on $\backslash p\{M\}$ characters. A regex match on s consults only characters of s , and the order of the alternatives is unchanged. Since s has no $\backslash p\{M\}$ character, both regexes produce the same pre-tokens. *Merges and vocabulary.* R2 differs from the original tokenizer only by appended merges and the vocabulary entries they create, each of which contains a complete code point in U+0900–U+097F. A merge (a, b) applies only where a and b are adjacent symbols of a pre-token, so its output is a contiguous substring of that pre-token’s bytes. A vocabulary entry is emitted directly under `ignore_merges` only if it equals the whole pre-token. Neither can happen for a pre-token of s , because UTF-8 is self-synchronising and s contains no such code point. Each pre-token of s is therefore encoded with original merges and origi-

nal vocabulary entries only, exactly as by the original tokenizer. *Ids*. New ids are appended above the original ids, so all ids coincide. □

D Training and evaluation protocol

Data. We normalise to NFC, drop documents under 200 characters, remove exact duplicates, and drop every document that shares a whitespace 10-gram with FLORES-200 dev or devtest in Nepali or English. This also covers the Belebele passages, which are FLORES passages (Bandarkar et al., 2024), but not its questions or answers. Documents are stored one JSON-encoded document per line. A 30 MB held-out slice of each language is carved from the cleaned stream before training and never trained on. The 1 GB of Nepali used to learn merges is taken from the start of the training stream.

Continued pretraining. Documents are interleaved and separated by the model’s end-of-text token; no BOS token is inserted. Optimisation is identical across arms: sequence length 2,048, global batch 256 sequences (0.52M tokens), AdamW ($\beta = (0.9, 0.95)$), weight decay 0.1), gradient clipping at 1.0, 1% linear warm-up, then cosine decay to 10% of peak, bf16 autocast with fp32 master weights, on $4 \times \text{H200}$. Initialisation is deterministic and there is no dropout, so a second seed would change only the order of training windows and bf16 non-determinism. In the warm-up runs, stage 1 trains the whole tied embedding matrix, old rows included, at 10^{-3} on the first 10% of the stream; stage 2 is a main run started from the stage-1 model.

Evaluation. Every scored sequence is preceded by one context token (§6), identically for every arm of a model.

- **BPB** on the held-out native slices (the first 4 MB of each) and on FLORES-200 devtest. In the primary byte-matched version, each held-out document is split at whitespace into pieces of at most 2,000 bytes and every piece is scored independently, so every arm conditions on the same bytes. Token windows would give R2 more bytes of context per scored token. As a secondary metric, whole documents are scored with a sliding window of 2,048 tokens and stride 1,024.
- **Belebele** (Bandarkar et al., 2024) *npi_Deva* and *eng_Latn*, zero-shot, scoring each an-

	Nepali	English
documents read	170,790	215,961
dropped: under 200 chars	0	0
dropped: exact duplicate	0	69
dropped: FLORES 10-gram	0	16
training documents	166,300	209,713
held-out (MB)	30	30
CPT tokens per arm (millions), 1 GB + 1 GB		
Llama-3.2-1B R0	207	211
Llama-3.2-1B R1	176	211
Llama-3.2-1B R2	76	211
Qwen3-0.6B R0	359	215
Qwen3-0.6B R1	177	215
Qwen3-0.6B R2	76	215

Table 2: Corpus cleaning (FineWeb-2 *npi_Deva*; FineWeb-Edu; documents placed in the 30 MB held-out slice are not counted as training documents) and the token counts of each arm’s identical 2 GB training stream. English token counts differ across arms of a model by less than 0.012%, from the few English documents that contain Devanagari or combining marks.

swer string by its log-likelihood per byte given passage and question.

- **Translation:** FLORES-200 devtest Nepali↔English, 5-shot from FLORES dev, greedy decoding capped at 512 new tokens, first output line, chrF++.
- **Generation speed:** UTF-8 bytes generated per second in greedy continuation of FLORES dev Nepali sentences (batch 64, 128 new tokens).

The equal-compute checkpoints are scored on BPB only, on the first 1 MB of each held-out slice (567 Nepali pieces from 111 reconstructed documents).

E Data

FineWeb-2 is already deduplicated and filtered for length, which is why our own filters remove almost nothing.

F Full audit

Table 3 lists every distinct tokenizer. For each of the nine aliases we compared token ids with its partner on all 1,012 devtest sentences of English, Nepali, Hindi, Chinese and Arabic; all were identical. Kimi-K3 (requires executing remote code) and Mistral-Large-3 (degenerate Hugging Face load) are not included. The four tokenizers that do not round-trip belong to encoders and translation models; we keep them as reference points.

Tokenizer	Aliases	Regex	$ V $	ne/en	95% CI	Floor	Mark-init.	Lossless
IndicBERTv2	–	none	250,000	1.09	[1.08, 1.10]	–	6%	no
Arkios	–	mark-aware	65,536	1.12	[1.10, 1.13]	0.78	15%	yes
XLNet	–	none	250,002	1.12	[1.11, 1.14]	–	7%	no
NLLB-200	–	none	256,204	1.17	[1.16, 1.18]	–	6%	no
BLOOM	–	other	250,680	1.18	[1.16, 1.19]	–	17%	yes
mT5	–	none	250,100	1.46	[1.45, 1.47]	–	11%	no
Gemma-3	Gemma-4	none	262,145	1.53	[1.51, 1.54]	–	9%	yes
Sarvam-1	–	none	68,096	1.54	[1.52, 1.56]	–	14%	yes
o200k	gpt-oss	mark-aware	200,019	1.61	[1.60, 1.63]	0.80	26%	yes
Llama-4	–	mark-aware	201,135	1.83	[1.81, 1.85]	0.80	30%	yes
Mistral-NeMo	Sarvam-M	mark-aware	131,072	2.13	[2.11, 2.15]	0.79	33%	yes
Gemma-2	–	none	256,000	2.13	[2.11, 2.15]	–	19%	yes
Qwen3.5	–	mark-aware	248,077	2.20	[2.18, 2.23]	0.80	19%	yes
Llama-3	SmolLM3	letters-only	128,256	2.58	[2.55, 2.61]	2.19	55%	yes
TituLLM	–	mark-aware	170,497	2.60	[2.57, 2.63]	0.78	55%	yes
DeepSeek-V3	DeepSeek-V4	mark-aware	128,815	2.97	[2.93, 3.00]	0.78	38%	yes
GLM-5.3	–	letters-only	154,856	4.35	[4.29, 4.40]	2.19	33%	yes
Qwen2.5	Qwen3	letters-only	151,665	4.42	[4.36, 4.47]	2.17	32%	yes
Mistral-7B	–	none	32,768	4.44	[4.38, 4.50]	–	29%	yes
GLM-4.5	–	letters-only	151,365	4.76	[4.71, 4.82]	2.19	30%	yes
cl100k	Phi-4, OLMo-2, Granite-4.1	letters-only	100,277	4.76	[4.71, 4.82]	2.19	30%	yes
Falcon3	–	letters-only	131,072	5.07	[5.00, 5.14]	3.21	28%	yes
LFM2	–	letters-only	64,400	5.77	[5.70, 5.84]	2.15	24%	yes
GPT-2	–	letters-only	50,257	7.55	[7.46, 7.65]	3.23	19%	yes

Table 3: All 24 distinct tokenizers (33 released tokenizers) on FLORES-200 devtest. Aliases: tokenizers whose token ids are identical on every devtest sentence of five languages (Appendix F). Regex: letters-only, mark-aware, other, or none (no regex pre-tokenizer). ne/en: Nepali/English token premium with 95% paired-bootstrap interval. Floor: pre-token floor under the tokenizer’s own regex, a lower bound on the premium of any merge-based extension (letters-only and mark-aware tokenizers only). Mark-init.: share of Nepali tokens whose first character is a combining mark. Lossless: decoding reproduces the input on 200 Nepali sentences. Arkios is from [Regmi et al. \(2026a\)](#). Hugging Face repositories are listed in the released code.

Broken syllables. The share of Nepali tokens that begin with a combining mark is 55% for Llama-3, 32% for Qwen3 and 30% for cl100k. Mark-aware tokenizers show similar rates (26% for o200k, 55% for TituLLM, which keeps Llama-3’s merges), because BPE also splits within words.

G Premium decomposition

We write the premium as

$$\frac{|T(\text{ne})|}{|T(\text{en})|} = \underbrace{\frac{|P_{\text{rep}}(\text{ne})|}{|T(\text{en})|}}_{\text{base}} \cdot \underbrace{\frac{|P(\text{ne})|}{|P_{\text{rep}}(\text{ne})|}}_{\text{regex factor}} \cdot \underbrace{\frac{|T(\text{ne})|}{|P(\text{ne})|}}_{\text{vocabulary factor}},$$

where P is the tokenizer’s own pre-tokenizer and P_{rep} is the repaired Llama-3 pattern, applied to every tokenizer so that the numerator of the base term is the same for all. The base lies between 0.77 and 0.80 because English token counts barely differ. Using one pattern means that Qwen’s regex factor also includes its single-digit splitting. The regex factor is 0.98–1.02 for mark-aware tokenizers, 2.77–2.79 for the Llama-3, Qwen and cl100k pattern, and 4.06 for the GPT-2 and Falcon3 patterns, which put every run of marks in a pre-token

of its own. The decomposition is an accounting identity and does not attribute cause. The regex factor is the room that a better regex creates, which new merges must then fill.

Table 4 shows the factors. For Llama-3 the vocabulary factor is 1.18, the lowest in the audit: its 128k vocabulary already nearly covers Nepali pre-tokens. For Qwen and cl100k both factors matter. Qwen3.5’s word branch is our repaired class with only the leading-character class left unchanged; [Land \(2026\)](#) reported its change for Hindi.

H Cross-script sweep

For each language we used the FineWeb-2 *test* split, normalised to NFC, with documents that share a whitespace 10-gram with that language’s FLORES-200 dev or devtest removed (for Thai we also checked 50-character overlaps and found none). The merge-learning text ranges from 25 MB (Amharic) to 361 MB (Bengali), less than the 1 GB used for Nepali, so for languages where R1 ran out of merges we cannot fully separate the floor from data scarcity; Sinhala and Amharic R2 may also be limited by data.

Tokenizer	Premium	Base	Regex	Vocab.
GPT-2	7.55	0.80	4.06	2.34
LFM2	5.77	0.78	2.77	2.68
c1100k (+3)	4.76	0.79	2.77	2.17
Qwen2.5/3	4.42	0.78	2.79	2.03
GLM-5.3	4.35	0.79	2.77	1.98
Llama-3	2.58	0.79	2.77	1.18
DeepSeek-V3/V4	2.97	0.80	0.98	3.79
Qwen3.5	2.20	0.78	1.02	2.76
Llama-4	1.83	0.80	1.00	2.30
o200k (+1)	1.61	0.80	1.00	2.01
Arkios	1.12	0.79	0.99	1.43

Table 4: Premium = base $\times$ regex factor $\times$ vocabulary factor, on FLORES-200 devtest, for 11 of the 15 letters-only and mark-aware tokenizers in the audit (top: letters-only; bottom: mark-aware). Merge-based extension can only lower the last column, and not below 1.

Lang.	Marks	Llama-3			Qwen3		
		R0	R1	R2	R0	R1	R2
Hindi	31%	2.52	2.38	1.17	4.42	2.36	1.16
Bengali	32%	5.84	2.29	1.01	5.03	2.26	1.00
Tamil	35%	7.65	2.67	1.02	6.11	2.64	1.02
Telugu	36%	8.29	2.45	1.02	6.99	2.43	1.01
Sinhala	28%	8.63	2.30	1.20	6.88	2.33	1.24
Burmese	50%	11.62	3.02	1.21	8.93	2.99	1.11
Thai [†]	21%	2.19	1.49	0.95	2.55	1.48	0.88
Amharic	0%	7.59	0.98	0.98	4.05	0.98	0.98
Korean	0%	1.49	0.98	0.98	1.63	0.98	0.98

Table 5: Language/English premium on FLORES-200 devtest, $K=32,000$. Marks: share of NFC code points that are combining marks; Amharic and Korean have none (controls). [†]Thai is written without spaces, so its pre-tokens are long and R1 is not yet at its floor. For R1, merge learning stopped below 32,000 for Tamil, Telugu, Sinhala, Burmese and Hindi (Llama) (22,325–31,815 merges), because no further merge was available under the letters-only regex.

A relaxed merge filter. The Nepali filter keeps a new merge only if its output contains a complete Devanagari character. Scripts that the base vocabularies cover poorly have characters split into byte pieces, and this filter would drop the first byte-level merge of each such character and every merge built on it (for Sinhala R2 on Llama-3, only 13,530 of 32,000 merges survived). For this sweep we therefore also keep merges whose output is a byte piece of a target-script character. Such merges can fire on other scripts that share those bytes: the lead byte 0xE0 begins every code point from U+0800 to U+0FFF, which covers all Indic scripts and Thai, and a continuation-byte suffix can end code points in other blocks. They can-

not fire on ASCII text, which contains no such bytes, and we verified English token ids on all 1,012 sentences for every tokenizer. Proposition 1 does not cover these tokenizers, and we did not measure their effect on other non-ASCII scripts. Hindi shares the Nepali script, and Nepali merges also change Hindi tokenization.

Thai and Burmese. Both are written without spaces between words, so a repaired pre-token can span several words. In practice no R2 token spans a space in either language. In Thai, 28% (Llama-3) and 25% (Qwen3) of R2’s Thai-script tokens cover more than one dictionary word (PyThaiNLP newmm segmentation), fewer than under R1 (33% and 31%). In Burmese, R2’s tokens average 2.2 and 1.8 syllables, against 1.3 under R1 and 5.1 syllables per space-delimited chunk.

I Released extensions

We did not follow a fixed search protocol. Candidates came from the papers in §2 and from manual searches of Hugging Face for Llama-3, Qwen and LFM2 derivatives in Indic and Southeast Asian languages. We kept releases whose tokenizer adds merges to a letters-only base. After inspection we excluded five releases that keep the base Llama-3 tokenizer unchanged and three that do not extend a letters-only base. The Llama-3 Hindi extension of Nanda (Choudhury et al., 2025) is gated and could not be measured, and we found no public tokenizer for Nag et al. (2024) or Purason et al. (2026). The Yamaguchi et al. extensions re-order the base merges and are not English-invariant (English tokens rise by 6.5% for Llama-3 Telugu and 7.8% for Qwen3 Telugu), so Table 6 compares tokens with pre-tokens of the target text directly. Two releases do not obey the letters-only floor: TituLLM (Nahin et al., 2025) changes the regex, and SinLlama (Aravinda et al., 2025) registers 11,080 strings as added tokens and reaches 0.54 times its letters-only floor.

J Additional tokenizer arms

Regex alone. Repairing the regex without adding merges (R2, $K=0$) gives 2.61 for Llama-3 (from 2.58) and 4.42 for Qwen3. Under the repaired regex the old merges still tokenize Nepali words in old pieces; the repair only removes the bound that new merges would otherwise hit.

Base	Lang.	Tokens/floor	Gap closed	Room
Llama-3-8B	te	1.016	99.3%	3.16×
Llama-3-8B	si	1.021	99.3%	2.42×
Llama-3-8B	my	1.013	99.5%	5.87×
Llama-3.1-8B	ta	1.018	99.1%	3.58×
Llama-3.1-8B	bn	1.020	98.7%	2.74×
Llama-3.1-8B	gu	1.026	98.9%	2.52×
Qwen2.5-7B	si	1.019	99.1%	2.40×
Qwen2.5-7B	my	1.036	98.2%	5.72×
Qwen3-14B	bn	1.019	98.5%	2.71×
Qwen3-14B	te	1.027	98.6%	3.11×
LFM2 (2.5)	hi	1.068	95.5%	2.24×
LFM2 (2.5)	ne	1.160	90.5%	2.77×
LFM2 (2.5)	bn	1.243	92.4%	2.74×
LFM2 (2.5)	th	1.819	87.9%	5.57×
Llama-3.1-8B	am	1.573	93.8%	1.00×

Table 6: Released merge-based extensions that keep the letters-only regex, on FLORES-200 devtest of the target language. Tokens/floor: extended-tokenizer tokens divided by pre-tokens under its own regex (at least 1 by construction). Gap closed: share of the base-to-floor gap that the extension removed. Room: pre-tokens under the letters-only regex divided by pre-tokens under the repaired regex. First block: Yamaguchi et al. (2026); second: Yamaguchi et al. (2025), with the two Qwen3-14B rows from a later release by the same authors; third: LFM2.5 (Smith et al., 2026); last: the Amharic release of Yamaguchi et al. (2025), a script with no combining marks (control).

Added tokens. Table 7 gives the added-token arm of §3. Default matching reaches almost R2’s compression and the lowest mark-initial rate, but splits many words at arbitrary points: a fifth of the added-token matches start inside a word and more than a quarter end inside one. With whole-word matching, few strings ever match and the premium returns close to the original.

Broken syllables. Counting only tokens that contain Devanagari, the share that begin with a combining mark rises from 57% (Llama-3) and 33% (Qwen3) to 67% for both under R1, and falls to 15% and 12% under R2. The letters-only regex leaves few distinct Nepali pre-tokens, so 5k to 30k merges cover nearly all frequent ones, which is why R1 saturates early.

K All continued-pretraining runs

BOS diagnostic. On 40 held-out Nepali documents (first 1,500 characters), Llama BPB after BOS and after the end-of-text token are 0.720 and 1.865 for the base model, 0.441 and 0.438 for R0 of the main runs, and 0.758 and 0.438 for R0 of the

Tokenizer	Premium		Mark-initial	
	Llama-3	Qwen3	Llama-3	Qwen3
R0	2.583	4.415	55%	32%
R1, $K=32k$	2.204	2.187	65%	64%
R2, $K=32k$	1.014	1.014	13%	11%
R2 strings, added	1.025	1.009	7%	7%
whole words only	2.486	4.219	54%	32%
Added-token share	77%	83%		
Starts mid-word	21%	21%		
Ends mid-word	28%	28%		

Table 7: Added tokens on the unchanged letters-only regex, FLORES-200 devtest. We register the 31,753 of R2’s 32,000 new strings that decode to whole Devanagari text (with at most one leading space) as added tokens. Premium: Nepali tokens / English tokens. Mark-initial: share of Nepali tokens that start with a combining mark. The bottom rows describe default matching, which ignores word boundaries: the share of tokens that are added tokens, and the share of added-token matches that start or end inside a word. No sentence without a Devanagari character changes its token ids (191,460 FLORES-200 sentences, both models, both matching modes).

letters-only $\backslash p\{L\}+$ न े प ा ल क ो स र क ा र

repaired $[\backslash p\{L\}\backslash p\{M\}]+$ नेपालको सरकार

Figure 4: Pre-tokens of *nepālko sarkār* (“government of Nepal”) under the letters-only regex of Llama-3 and Qwen and under the repaired regex. Dotted circles mark pre-tokens that begin with a dependent vowel sign.

warm-up runs. The norm of the BOS embedding is 0.863, 0.860 and 0.896 respectively (mean row norm 0.98–0.99). The base model, pretrained with BOS at sequence starts, scores far worse without it, while the continued-pretraining models, trained on fixed-length windows that start at arbitrary tokens, do not need it. We did not test the mechanism further.

Secondary metrics. Belebele is near chance for every model. Translation scores are erratic: Qwen R2 scores 3.9 chrF++ from English to Nepali, below the untrained base model (6.8). We take the first output line as the hypothesis, which is empty when a model starts with a newline, and we did not measure how often that happens. Qwen’s FLORES BPB is the same for R1 and R2 (0.689), although R2 is worse on held-out native text. We draw no conclusions from these metrics.

Model	Runs	Arm	Tokens	held-out ne BPB↓		en BPB↓	FLORES	Belebele	chrF++		Gen.
				reg.	sep.	reg.	ne BPB	ne acc.	ne→en	en→ne	B/s
Llama-3.2-1B	none	base	0	0.766	n/a	0.788	0.863	0.277	32.4	10.8	5327
	main	R0	417M	0.474	0.471	0.799	0.611	0.287	32.3	19.8	5147
	main	R1	386M	0.534	0.483	0.803	0.648	0.268	27.2	10.4	6354
	main	R2	286M	0.539	0.510	0.799	0.671	0.266	27.8	12.3	9398
	warm-up	R0	417M	0.779	0.471	0.850	0.864	0.279	8.8	9.1	5349
	warm-up	R1	386M	0.808	0.480	0.854	0.904	0.271	9.9	4.2	7521
	warm-up	R2	286M	0.527	0.501	0.840	0.652	0.283	16.2	15.4	11078
Qwen3-0.6B	none	base	0	0.852	0.852	0.873	0.945	0.290	26.0	6.8	2186
	main	R0	574M	0.501	0.501	0.864	0.631	0.270	20.1	13.6	3089
	main	R1	392M	0.561	0.561	0.864	0.689	0.253	11.6	6.9	4972
	main	R2	292M	0.570	0.570	0.862	0.688	0.266	17.0	3.9	7454

Table 8: Every continued-pretraining run (one run per arm). Main runs: 1 GB of Nepali and 1 GB of English. Warm-up runs: the same, after training only the embedding matrix on the first 10% of the stream (Tokens excludes that stage). Held-out BPB is scored in pieces of about 2,000 bytes, after the registered prefix (reg.: beginning-of-sequence token for Llama) or the training separator (sep.: end-of-text token); for Qwen the two coincide. The base Llama model was never trained without its beginning-of-sequence token, so its separator score is not meaningful (n/a). All columns other than sep. use the registered prefix; for the Llama warm-up runs they are dominated by the scoring artifact of §7.2 and should not be compared with the main runs. Belebele: zero-shot accuracy on 900 items (chance 0.25); every value lies between 0.25 and 0.29 with 95% intervals of about ± 0.03 , so it cannot separate the arms. chrF++: 5-shot FLORES-200 devtest translation, single greedy pass, no intervals. Gen. B/s: UTF-8 bytes of Nepali generated per second, measured on GPUs shared with concurrent training runs; it is noisy (the base and R0 Qwen models share tokenizer and architecture yet differ by 41%) and we do not draw conclusions from it.

Measured generation speed. Generation was timed once per model, on GPUs that were also running training or evaluation jobs. The Qwen base model and Qwen R0 share tokenizer and architecture, yet differ by 41% in bytes per second, and the generated text has fewer bytes per token than FLORES (8.1 against 12.3 for Llama R2), so the measured speed also depends on what each model chose to write. Section 7.2 therefore reports the deterministic ratio of decoding steps instead.